

Nitin Lohan Ph.D.

AI Trainer & STEM Benchmark Developer | Mathematics LLM Evaluator | Scientific Data Analyst

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PROFESSIONAL SUMMARY

Applied Mathematics Ph.D. and AI Trainer / Evaluator specializing in LLM evaluation, model-output grading, RLHF / human-feedback data, STEM benchmark development and model-failure analysis. Experienced in designing high-difficulty mathematics and numerical-methods tasks, authoring rigorous reference solutions, validating AI outputs for correctness, and revising tasks across reviewer cycles to satisfy exact-match scoring. Backed by a strong scientific-computing foundation (expert Python, machine learning, numerical weather modeling with WRF/WRFDA, 3DVAR data assimilation and HPC workflows) and peer-reviewed technical writing across 7 journal articles.

CORE SKILLS

AI Training & LLM Evaluation	LLM evaluation and response grading; RLHF / human-feedback data; reward-model preference data; prompt and task design; red-teaming and model-breaking reasoning tasks; rubric and annotation-guideline design; hallucination, error and edge-case detection; reference-solution writing; inter-rater consistency; citation and numerical verification; dataset quality assurance
Benchmark & Task Design	STEM benchmark development; SciCode-style multi-step tasks; exact-match and rubric-based scoring; discriminating test-case engineering; operator and scoring-logic redesign; evaluation-harness debugging; reproducibility validation
Mathematics	Numerical linear algebra; partial differential equations; numerical analysis and interpolation; probability, statistics and extreme-value theory; optimization; finite-difference and finite-element methods
Programming & Data	Python (expert): NumPy, SciPy, pandas, xarray, Matplotlib, Cartopy, wrf-python; CDO; machine learning; reproducible, research-grade scientific code
Scientific Computing	WRF / WRFDA; 3DVAR data assimilation; HPC (SLURM/PBS); parallel computing; model automation and scheduling; ERA5, IMDAA, IMD, GPM and INSAT datasets
Tools	LaTeX; Git/GitHub; JupyterLab; VS Code; Cursor; Claude Code; Linux; Windows

PROFESSIONAL EXPERIENCE

AI Trainer / Evaluator & STEM Benchmark Developer Sep 2025 – Present
Turing & Mercor · Remote

- Designed 100+ high-difficulty mathematics and numerical-methods benchmark tasks to evaluate and improve advanced AI/LLM reasoning.
- Authored rigorous, step-by-step reference solutions with explicit assumptions, derivations, unit consistency and verified final answers.
- Evaluated and graded AI model outputs across multiple model families for mathematical correctness, reasoning depth, hallucinations, ambiguity, formatting and answer consistency.
- Revised benchmark tasks through 3–4 reviewer cycles per task, redesigning operators and scoring logic to satisfy exact-match evaluation.
- Established a strict pre-submission verification discipline: numerical outputs computed and citations confirmed against source PDFs before delivery, reducing reviewer-rejection errors.
- Built WENO-type reconstruction and SciCode-style multi-step Python benchmarks tied to numerical weather-prediction methods.

Assistant Professor (Mathematics) Jul 2025 – Present
G.L. Bajaj Institute of Technology and Management · Greater Noida

- Teach undergraduate engineering mathematics to 100+ students; develop course material, assessments and problem sets.
- Mentor students in applied mathematics, numerical methods and scientific computing with Python.

Gautam Buddha University · Greater Noida

- Conducted diagnostic and numerical-modeling studies of heavy-precipitation and cloudburst events over the Indian region.
- Configured and ran WRF/WRFDA simulations on HPC clusters; applied 3DVAR data assimilation to high-impact case studies.
- Analyzed ERA5, IMDAA, IMD, GPM and INSAT datasets for extreme-rainfall and monsoon-system diagnostics.
- Published 7 peer-reviewed journal articles and 3 conference papers / book chapters; delivered 3 national and international conference presentations.

SELECTED AI / BENCHMARK PROJECTS

STEM Benchmark Development for LLM Evaluation · Mercor / Turing

- Developed challenging mathematics and numerical-analysis tasks across 15+ benchmark projects, exposing model weaknesses in reasoning, derivation and numerical consistency.
- Engineered discriminating test cases and operator redesigns (e.g. monotone Hermite interpolation, inequality-based estimators) to pass exact-match scoring.
- Verified all numerical outputs and citations against source PDFs before producing copy-paste-ready deliverable files.

SciCode-Style Multi-Step Python Benchmarks

- Built 25+ reproducible benchmark notebooks spanning FEM, phase-field fracture, poroelasticity and stochastic-flow problems from recent arXiv papers.
- Diagnosed harness artifacts versus genuine model failures and recalibrated evaluation pipelines for fair scoring.
- Delivered complete, research-grade code with publication-quality numerical reasoning and documentation.

EDUCATION

Ph.D. in Applied Mathematics, Gautam Buddha University, Greater Noida 2020 – 2026 (defended Jan 2026)

Thesis: *Diagnostic and Numerical Modeling Study of Heavy Precipitation Events over Indian Region* · Supervisor: Dr. Sushil Kumar · CGPA 8.29

M.Sc. in Mathematics, DAV College, Panjab University, Chandigarh 2016 – 2018

B.Sc. (Physics, Chemistry, Mathematics), Govt. PG College Jind, Kurukshetra University 2013 – 2016

SELECTED PUBLICATIONS

- **Lohan, N.**, Pattanaik, D. R., Kumar, S., & Giri, R. K. (2025). Diagnostic study of monsoon-linked rainfall events: Unraveling the role of monsoon low pressure systems across the Indian subcontinent. *Earth Systems and Environment*. doi:10.1007/s41748-025-00706-y
- **Lohan, N.**, Routray, A., Kumar, R., Mahala, B. K., & Kumar, S. (2025). Validation and diagnostic study of cloudburst events over the Himalayan region using IMDAA and ERA5. *Geomatics, Natural Hazards and Risk*, 16(1), 2446589. doi:10.1080/19475705.2024.2446589
- **Lohan, N.**, Kumar, S., Singh, V., Gupta, R. P., & Tiwari, G. (2025). Analyzing an extreme rainfall event in Himachal Pradesh, India, to contribute to sustainable development. *Sustainability*, 17(5), 2115. doi:10.3390/su17052115

Full list (7 journal articles, 3 conference papers / book chapters): [Google Scholar](#)

QUALIFICATIONS, COLLABORATIONS & TRAINING

- **CSIR-UGC NET** (Mathematical Sciences), qualified 2019 and 2022 · **GATE Mathematics**, qualified 2022
- **Research collaborations** with IMD (Dr. R. K. Giri, Dr. D. R. Pattanaik), NCMRWF (Dr. Ashish Routray) and IITM Pune on HPC-based modeling and 3DVAR data assimilation
- **Training:** IMD-Pune Induction Training Program on Meteorology (2024); NIDM Disaster Risk Reduction Programme (2023) · **Presentations:** 2 oral (TROPNET 2023, ICAIM 2023), 1 poster (TROPNET 2022)
- **Languages:** Hindi (native), English (fluent)